Discussion 8

Def 1: Uniform Discrete Random variable

We say $X \sim DisUnif(\{a, a+1, \cdots, b\})$ when Random variable X can only takes discrete values $a, a+1, \cdots, b$ and they are all equally likely.

$$
\begin{aligned}
p_X(x) =& \frac{1}{b-a+1};\ \textit{where } x = a, a+1, \cdots, b \\
E[X] =& \frac{a+b}{2} \\
Var(X) =& \frac{(b-a+1)^2 - 1}{12}
\end{aligned}
$$

Eg1. A fair die is tossed twice, independently. Let the random variable Z be defined as the number of dots seen on the first toss minus the number of dots seen on the second toss. Find the expected value of Z^2.

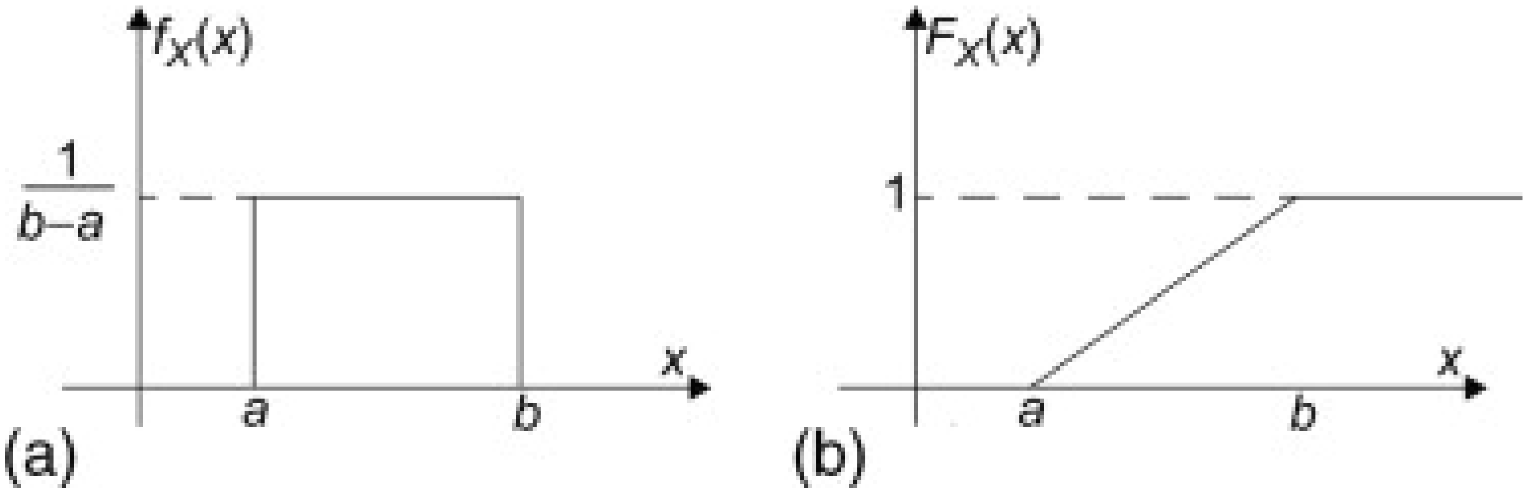

Figure 1: probability density function and cumulative distribution function of a uniformly distributed random variable

Def 2: Continuous Uniform Distribution $T \sim Uniform(a, b)$

The random variable X is uniformly distributed over the interval (a,b) when its probability density function is given by

$$f_T(t) = \begin{cases} \dfrac{1}{b-a} & \text{if } a < t < b \\ 0 & \text{otherwise} \end{cases}$$

It can be shown that

$$F_T(t) = \begin{cases} 0 & \text{if } t < a \\ \dfrac{t-a}{b-a} & \text{if } a \leq t < b \\ 1 & t \geq b \end{cases}$$

$$E[T] = \frac{a+b}{2}$$

$$Var(T) = \frac{(b-a)^2}{12}$$

Eg 2. Buses arrive at a specified stop at 15-minute intervals starting at 7 A.M. That is, they arrive at 7, 7:15, 7:30, 7:45, and so on. If a passenger arrives at the stop at a time that is uniformly distributed between 7 and 7:30, find the probability that he waits

(a) less than 5 minutes for bus

(b) more than 10 minutes for a bus

3. Let $X_1, X_2, \cdots, X_{80}$ be a random sample of size $n = 80$ from a distribution with probability density function

$$f(x) = \begin{cases} 6x(1-x) & 0 < x < 1 \\ \\ 0 & \text{otherwise} \end{cases}$$

Calculate $P(0.48 < \bar{X} < 0.52)$. Hint: Begin by calculating μ and σ^2.

4. Use the Central limit theorem to approximate the following probabilities.

(a) Suppose we roll a fair 7-sided die until we get 50 seven's. What is the probability it takes at least 500 rolls until this happens? (use continuity correction for this part.)

(b) Let X be the sum of 500 real numbers, and Y be the same sum, but with each number rounded to the nearest integer before summing. If the fractions rounded off are independent and each one is uniformly distributed over (-0.1, 0.1), use the Central Limit Theorem to estimate the probability that $P(|X - Y| > 30)$.